

Stock Price Prediction Using LSTM Neural Network: A Case Study Using Safaricom Stocks

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Introduction & Problem Statement

Stock price prediction is a critical challenge for investors. Traditional methods often fail to capture non-linear patterns in stock prices. This study proposes a multivariate Long-Short Term Memory (LSTM) model using technical indicators and lagged prices to enhance predictive accuracy.

The Data

The dataset consists of Safaricom PLC (SCOM) stock data from 2010 to December 2024, including Open, Close, High, Low, and Volume.

Date	Open	High	Low	Close	Volume
2010-01-04	4.55	4.55	4.55	4.55	2,802,800
2010-01-05	4.55	4.60	4.60	4.60	1,965,900
2010-01-06	4.60	4.80	4.80	4.80	3,856,500

Table 1: Sample stock data from Safaricom PLC

Feature Selection

- 44 total features consisting of technical indicators and up to 5 days lagged prices were generated.
- Recursive Feature Elimination (RFE) was used to select the most relevant features.
- We found that 20 features were the most optimal as they produced the lowest RMSE.

Number of Features	Loss (RMSE)
5	1.5082
10	0.7986
15	0.7476
20	0.7354
21	0.7377
22	0.7472
25	0.8992

Table 2: Loss (RMSE) for Different Numbers of Features

The LSTM Model

- LSTM is a type of Neural Network that can learn long-term dependencies.
- Its distinguishing feature is the memory cells contained within the hidden layers, allowing it to retain previous data and remove unimportant data.

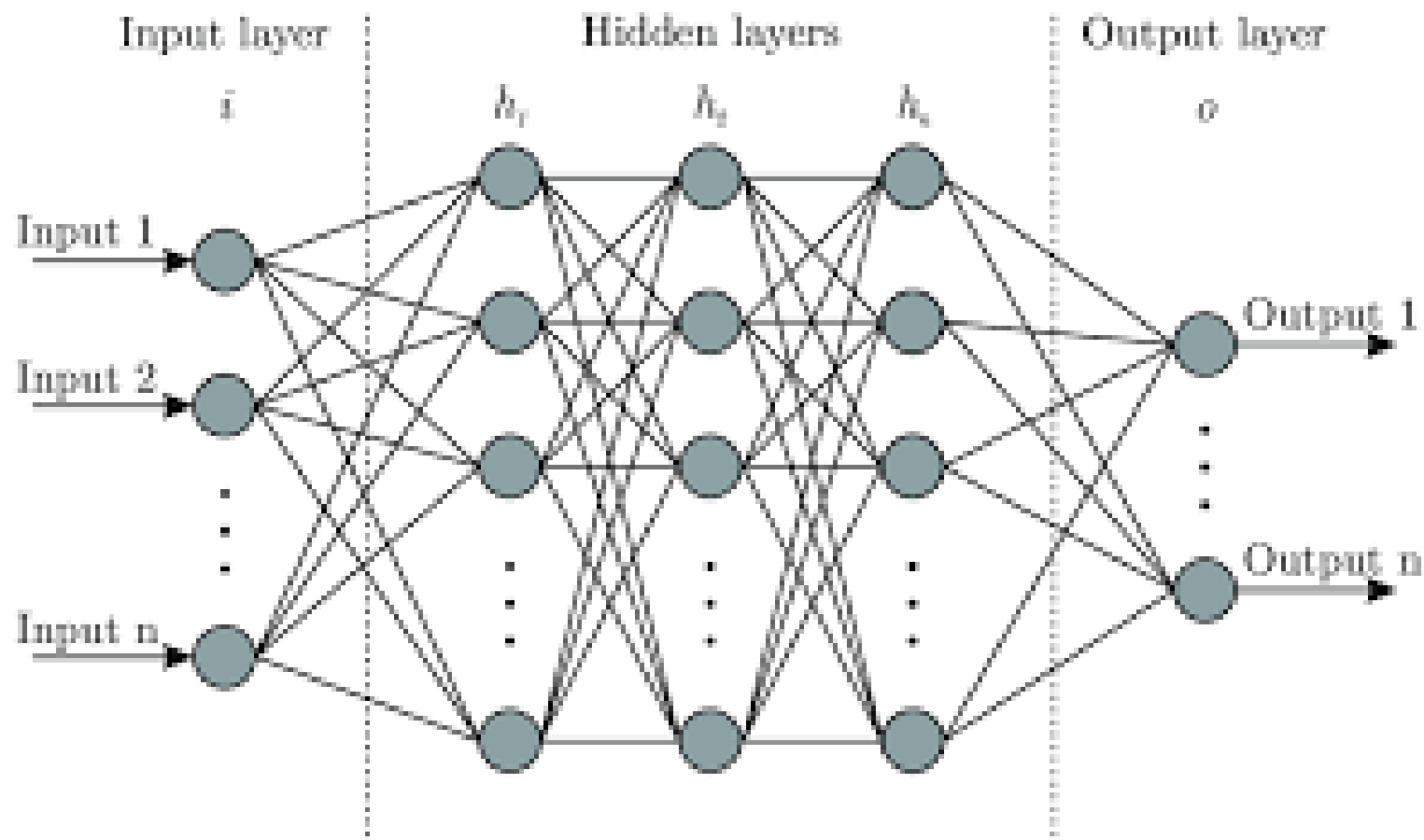


Figure 1: Neural Network Design

- A multivariate LSTM model was developed using 20 features selected via Recursive Feature Elimination (RFE) as inputs. We compare two model types:
 - Arbitrary LSTM:** Basic model with arbitrarily selected hyperparameters.
 - Tuned LSTM:** Fine-tuned model with optimized hyperparameters.
- Performance was assessed using MSE, RMSE, MAE, MAPE, and R-squared metrics

Model Performance

- The tuned LSTM outperformed the arbitrary model in all metrics.
- While both models struggled with sudden price shocks, the tuned LSTM was able to predict the general trend more accurately than the arbitrary LSTM.

Metric	Tuned LSTM	Arbitrary LSTM
MSE	9.3109	21.2153
RMSE	3.0514	4.6060
MAE	2.2803	3.4543
MAPE	0.1199	0.1763
R-squared	0.7562	0.4445

Table 3: Comparison of model performance metrics

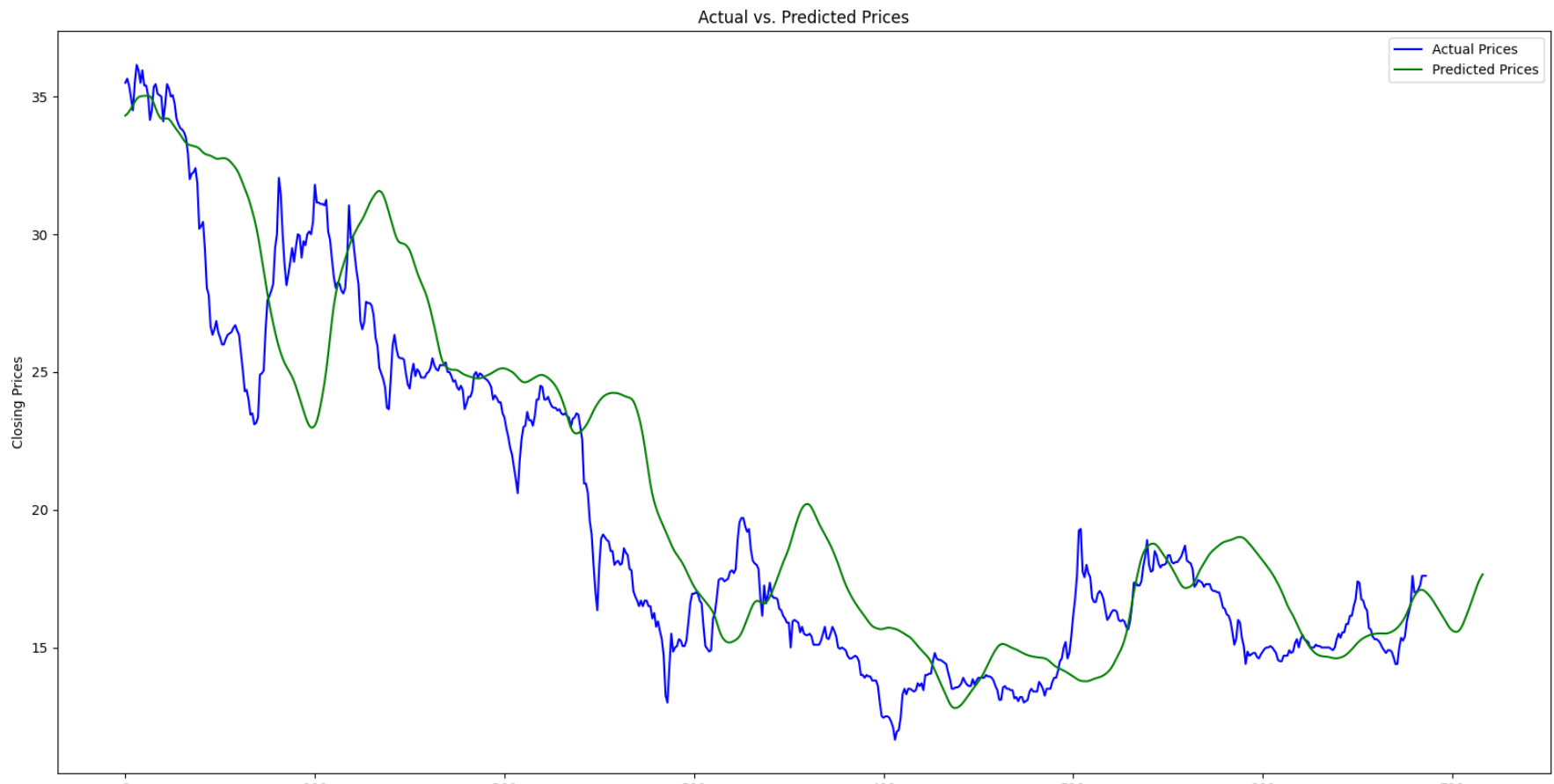


Figure 2: Forecasted stock prices using the tuned LSTM model

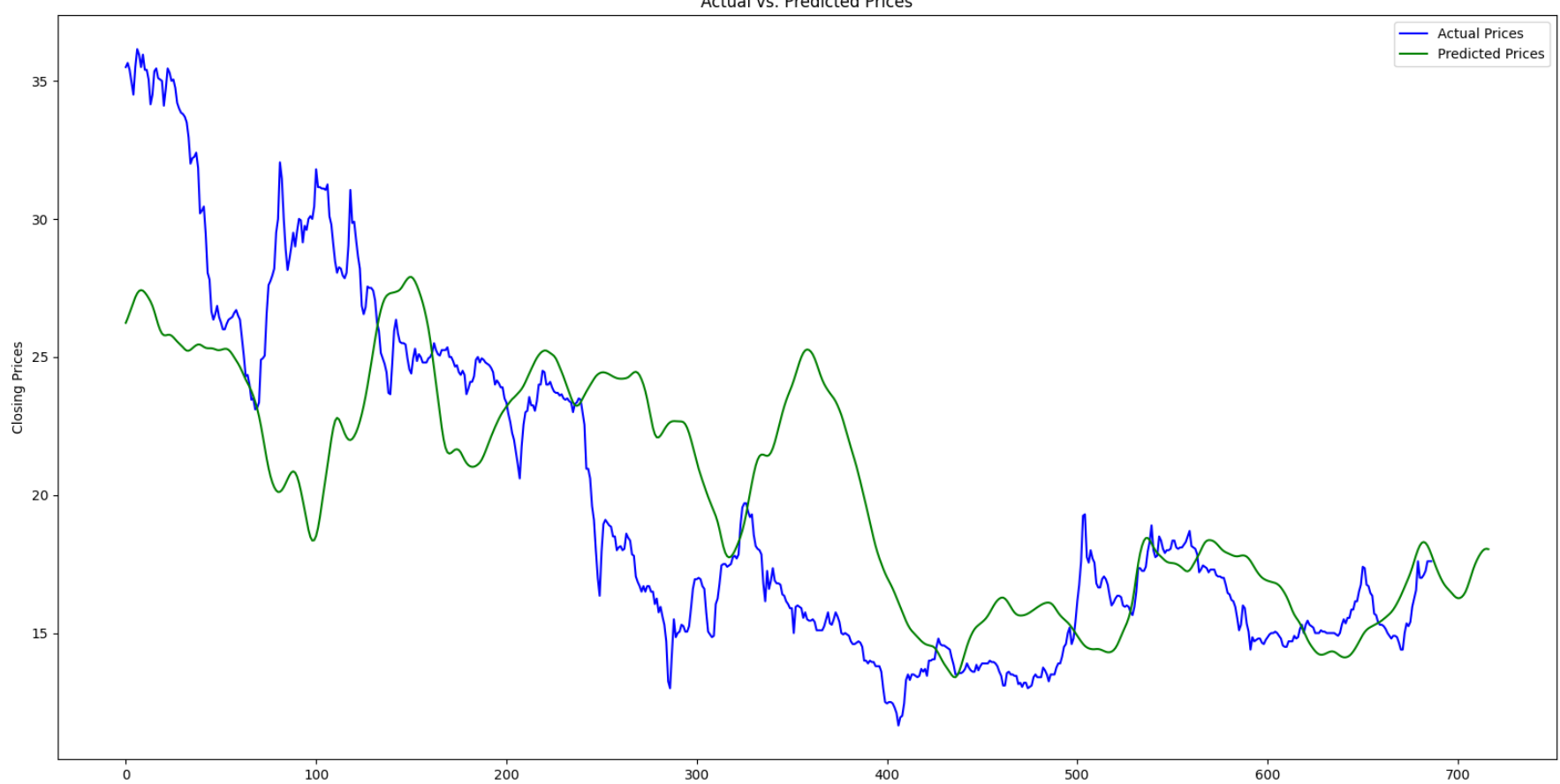


Figure 3: Forecasted stock prices using the arbitrary LSTM model

Loss Function

- The loss function is a metric used to improve model performance during training by minimizing the error between predicted and actual values.

- We implemented the Mean Squared Error (MSE) loss function for both the training and validation loss.

- Training Loss:** Measures the error on the training data during each iteration of model training (per epoch)
- Validation Loss:** Measures the error on a separate validation dataset that the model has not seen during training.

$$MSE = \frac{1}{N} \sum_{t=1}^N (P_t - \hat{P}_t)^2$$

- The tuned LSTM seems to have generalized well to the data while the arbitrary LSTM suffers from high variance and overfitting.

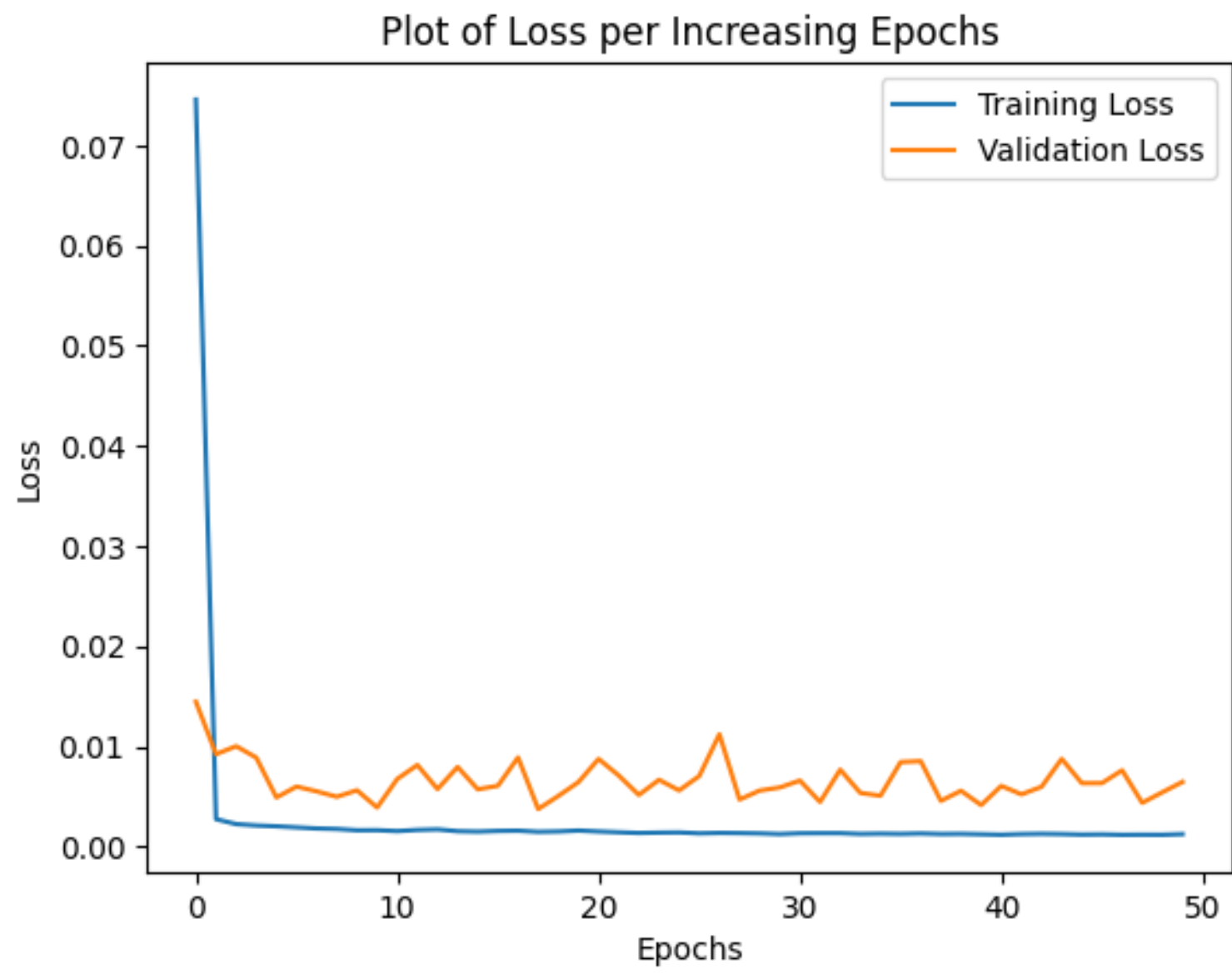


Figure 4: Tuned LSTM Loss Function During Training

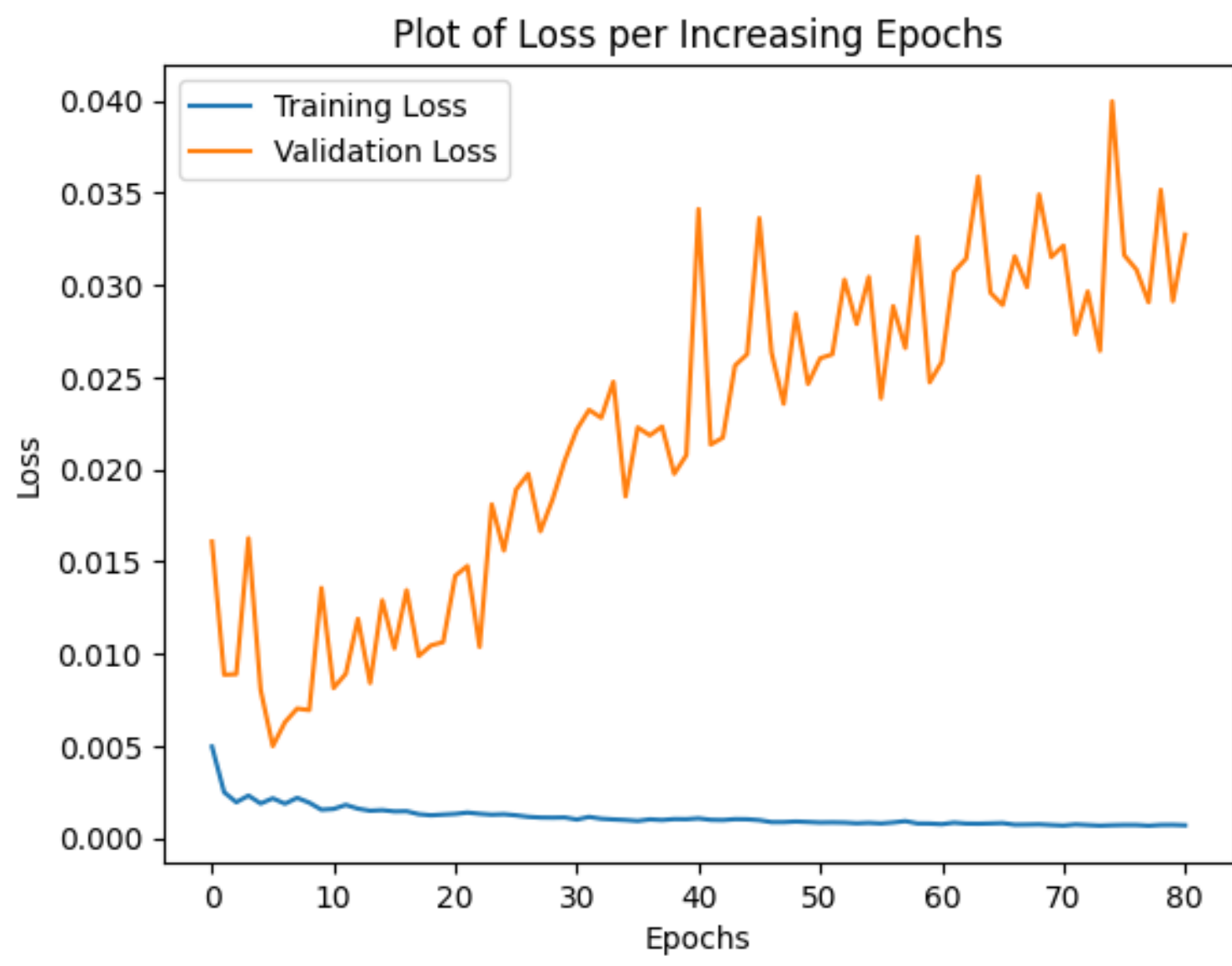


Figure 5: Arbitrary LSTM Loss Function During Training

Conclusion

- A well-constructed LSTM can significantly improve short-term stock price predictions. The tuned model outperformed the arbitrary LSTM, emphasizing the importance of hyperparameter tuning.
- Future studies could integrate additional variables such as market sentiment for improved accuracy.